

ALLO

Gemini Live Latency, Barge-in and Realism

Fact-checked implementation specification for Codex

REPOSITORY	TheHighBrid/Allo
AUDITED REVISION	main @ f31166fc3239751b67f99d0bd256193c341e6224
RESEARCH INPUT	research kenza delay_260729_041530.docx (278 pages)
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FINAL VERDICT

The uploaded research contains a strong technical core, but it is heavily duplicated and mixes verified guidance with contradictory, unsupported, outdated, or repo-irrelevant advice. This document keeps only the useful facts, corrects the false claims, audits the current Allo code, and gives Codex an ordered, testable implementation plan.

Deliverable goal: faster response onset, real interruption behavior, stable long calls, and no cargo-cult optimizations.

0. Codex execution directive

READ THIS FIRST

Implement the P0 and P1 work in this specification against the audited Allo revision. Preserve the current visual UI, Kensa identity, memory data model, and ElevenLabs behavior. Do not implement any item listed in Section 2 as REMOVE or DO NOT DO. Keep commits small, run the required checks, and report measured before/after results.

Required branch and commit sequence

1. Create branch: codex/gemini-live-latency-v2
2. Commit 1: perf(audio): decouple websocket receive from playback
3. Commit 2: feat(call): add route-aware full duplex and stream lifecycle
4. Commit 3: feat(telemetry): add turn timing and Gemini usage metrics
5. Commit 4: fix(session): improve GoAway handling and transcript turn assembly
6. Commit 5: security(live): add optional ephemeral-token mode for distributed builds

Non-negotiable constraints

- Keep raw WebSocket messages as JSON. PCM bytes must remain Base64 in the JSON Blob data field.
- Keep Gemini 3.1 Flash Live as the default model.
- Keep automatic VAD for the first implementation pass. Do not ship manual endpoint control yet.
- Keep 40 ms microphone packets as the default. Add 20 ms only as a controlled experiment.
- Do not add fake breathing, filler audio, artificial carrier jitter, telephone EQ, or random sound overlays.
- Do not send audioStreamEnd after every utterance. Send it only when the microphone stream is intentionally paused, such as mute.
- Do not add proactive audio or affective dialogue configuration to Gemini 3.1.
- Do not add a forced 8-12 minute session refresh. Use context compression, session resumption, and GoAway correctly.
- Do not log raw audio or private transcript text in latency telemetry.
- Keep minSdk 26 and use only existing dependencies unless a new dependency is clearly justified.

Definition of done

- ./gradlew test passes.
- ./gradlew assembleDebug passes.
- The WebSocket callback never performs blocking AudioTrack writes.
- Earpiece and headset routes allow the user to interrupt model speech.
- Speakerphone remains echo-safe and does not self-interrupt repeatedly.
- Mute sends one audioStreamEnd event per mute transition.
- Usage metadata and per-turn latency metrics are captured without transcript/audio content.
- A 10-minute and a 30-minute call survive connection rollover without audible stale audio.

1. Executive verdict

The research correctly identifies the main latency chain: endpoint detection, microphone buffering, network and model time-to-first-audio, and playback buffering. It also correctly recommends small PCM chunks, automatic VAD tuning, immediate streamed playback, interruption flushing, persistent sessions, context compression, and session resumption. [S2][S3][S6]

However, several repeated recommendations are wrong for the current Gemini Live WebSocket protocol or wrong for Gemini 3.1. Raw WebSocket integration uses JSON messages with Base64 Blob data, not arbitrary binary audio frames. Gemini 3.1 does not support proactive audio, affective dialogue, or asynchronous non-blocking function calling. audioStreamEnd is for a paused or closed microphone stream, not for every normal end of speech. [S1][S2][S4][S6]

The current Allo repository already contains many of the valid fixes: 40 ms PCM input, 16 kHz capture, 24 kHz output, a 600 ms automatic VAD window, a persistent WebSocket, all-part response parsing, context compression, session resumption, GoAway handling, and interruption-triggered playback flushing. [R1][R2][R3]

The largest remaining repo-specific latency risk is not the model configuration. It is the app's audio threading: AudioTrack.write is called directly from the OkHttp WebSocket callback. Android's default byte-array write is blocking, so the network receive path can stall behind speaker playback. The largest realism gap is route gating: built-in phone microphone routes stop sending audio while Gemini speaks, so normal earpiece calls cannot truly barge in. [R2][R3][R4][S8]

IMPLEMENTATION PRIORITY
1) move playback to a dedicated audio worker; 2) enable route-aware full duplex for earpiece and headsets; 3) add stream lifecycle and measurable telemetry; 4) harden long-session and transcript handling; 5) replace long-lived client API keys with ephemeral tokens before public distribution.

Priority map

Priority	Workstream	Current evidence	Expected result
P0	Playback worker	WebSocket receive callback currently performs blocking AudioTrack writes.	High latency and interruption risk.
P0	Route-aware barge-in	Only headset input bypasses the agentSpeaking gate.	Normal earpiece calls cannot interrupt.
P0	Mute stream lifecycle	Muted mic silently stops packets but sends no audioStreamEnd.	Server stream state can remain ambiguous.
P0	Turn telemetry	No local speech-end, first-audio, queue, underrun, route, or token metrics.	Tuning is guesswork.
P1	GoAway deadline	Current code waits a fixed four seconds instead of using timeLeft.	Reconnect timing can be unnecessarily risky.
P1	Transcript assembly	Fragments are appended as separate utterances.	Noisy memory and inaccurate call logs.
P1	Prompt and memory budget	Prompt plus memory briefing can become large and rigid.	Cost, first-turn latency, and style drift.
RELEASE	Ephemeral-token mode	Long-lived API key is compiled or stored on the device.	Unsafe for distributed builds.

2. Fact check of the uploaded research

This section is the pruning layer. It is intentionally decisive: KEEP means Codex may implement or preserve it; CORRECT means the underlying idea is useful but the original wording or parameter is wrong; REMOVE means it should not guide implementation.

Verdict	Claim	Fact-checked result	Implementation instruction
KEEP	Input/output format	16-bit little-endian mono PCM, 16 kHz input and 24 kHz output.	Official Live API specification. Current repo already matches it. [S2][S6][R2][R3]
KEEP	20-40 ms input chunks	Small chunks reduce client buffering. Current 40 ms packets are valid.	Do not force 20 ms. Keep 40 ms default and A/B 20 ms later. [S3][R1][R3]
KEEP	Automatic VAD 500-800 ms	Official guidance recommends 500-800 ms as the quality/latency range.	Current 600 ms is valid. Test 500 and 650, not 150-350. [S2][R1]
KEEP	Immediate streamed playback	Process each returned audio part and begin playback without waiting for turnComplete.	Keep. Fix the callback-thread blocking bug. [S2][S6]
KEEP	Flush on interruption	When serverContent.interrupted is true, stop and clear queued audio.	Current repo does this; retain and make the queue truly clearable. [S2][S3][R2][R4]
KEEP	Persistent session + resumption	Use one session, context compression, session resumption, and GoAway.	Current repo already implements the core. Improve deadline handling. [S3][S7][R2]
KEEP	Process every response part	Gemini 3.1 may put audio and transcript parts in one event.	Current for-loop is correct. [S1][R2]
KEEP	Minimal thinking	Gemini 3.1 defaults to minimal for lowest latency.	Current code correctly relies on the default. [S1][S2][R2]
KEEP	Usage metadata	Live server messages periodically include token totals.	Add parsing and telemetry. [S2][S6]
CORRECT	Base64 versus binary	PCM is raw in meaning, but the direct WebSocket protocol carries it inside JSON Blob data as Base64.	Do not replace current JSON/Base64 transport with arbitrary binary frames. [S4][S6][R2]
CORRECT	audioStreamEnd	It is used when the audio stream is paused or the microphone is turned off.	Send on mute transition, not at every normal end of speech. [S2][S6]
CORRECT	Direct client connection	Direct client-to-Gemini is latency-efficient.	For production, use ephemeral tokens, not a long-lived API key on device. [S5]
CORRECT	Manual VAD	Supported, but it increases implementation risk and can fragment speech.	Keep automatic VAD until telemetry proves it is the bottleneck. [S2]
REMOVE	150-350 ms default silence	Conflicts with official 500-800 ms recommendation and other sections of the research.	Do not ship as the default. [S2]
REMOVE	TCP_NODELAY and raw binary advice	Not part of the documented Live JSON WebSocket integration and not an actionable OkHttp tuning knob here.	Remove.
REMOVE	Dummy audio or text warm-up	No official basis and it can create unwanted model state, cost, or output.	Connect when the call begins; do not send fake input.
REMOVE	Forced 8-12 minute refresh	Session resumption and compression are the supported mechanisms.	Do not deliberately discard a healthy session. [S3][S7]
REMOVE	Proactive/affective config on 3.1	Gemini 3.1 does not support either feature.	Do not add these fields. [S1][S2]
REMOVE	Non-blocking tools on 3.1	Gemini 3.1 function calling is synchronous only; Allo does not need tools in the live speech path.	Remove from this workstream. [S1][S2]
REMOVE	Priority inference tier	Not a required or repo-applicable Gemini Developer API fix in the supplied architecture.	Remove unless a future Vertex deployment explicitly supports it.

Verdict	Claim	Fact-checked result	Implementation instruction
REMOVE	Fake breathing, filler clips, phone EQ	These effects are not needed and can reduce intelligibility, echo cancellation, and authenticity.	Use native prosody and concise behavior rules instead.
REMOVE	Fixed temperature/top-p recipe	No official evidence that one numeric pair produces lower latency or realism.	Keep current values until controlled listening tests show a benefit.
REMOVE	Packet-loss metric requirement	The app layer does not directly expose reliable packet loss for this WSS flow.	Log network type, socket queue, reconnects, and timing instead.

3. Current Allo repository audit

Audited repository: TheHighBrid/Allo, main commit f31166fc3239751b67f99d0bd256193c341e6224.

3.1 What is already correct

Central tuning constants: GeminiLiveTuning.kt defines 40 ms input packets, 20 ms prefix padding, 600 ms silence, and a 100 ms output target. [R1]

Correct native audio rates: AudioRecord captures 16 kHz PCM16 mono and Gemini reports 24 kHz output to PcmPlayer. [R2][R3][R4]

Automatic VAD configuration: Gemini setup uses high start/end sensitivity with 20 ms prefix and 600 ms silence. [R2]

Persistent WSS session: One OkHttp WebSocket is kept per call with ping, no read timeout, and incremental realtimeInput. [R2]

All-part response handling: Every modelTurn part is iterated and every inlineData audio part is decoded. [R2]

Continuity: Context-window compression, session resumption updates, GoAway, and reconnect handling are present. [R2]

Interruption response: serverContent.interrupted calls listener.onInterrupted and CallViewModel flushes the player. [R2][R4]

Output-rate handoff: CallViewModel creates PcmPlayer using the provider-reported 24 kHz rate. [R4]

Prompt discipline: The Gemini delivery addendum favors short spoken turns and avoids assistant-style verbosity. [R4][R6]

3.2 Critical code-specific gaps

#	Gap	Repository evidence	Why it matters	Required correction
1	Blocking receive path	GeminiLiveProvider.handle decodes audio and immediately invokes listener.onAgentAudio. CallViewModel then calls player.write. PcmPlayer.write calls AudioTrack.write directly. [R2][R3][R4]	AudioTrack byte-array writes are blocking by default. The WebSocket callback can stall behind playback. [S8]	Move all speaker writes to a dedicated audio-priority worker with a bounded queue.
2	Earpiece is half duplex	MicRecorder sends user audio during agent speech only when routed input is a headset. Built-in mic, including normal earpiece calls, is gated. [R3]	Gemini cannot detect barge-in if it receives no microphone audio.	Allow full duplex for earpiece and headsets. Keep speakerphone conservative until echo-safe testing.
3	No real latency evidence	No per-turn timing object, no socket queue measurement, no playback queue duration, no underrun count, and usageMetadata is ignored. [R2]	The app cannot distinguish VAD delay from model delay or playback delay.	Add private, content-free telemetry before tuning constants.
4	Mute does not close stream	toggleMute only sets mic.muted. Mic capture continues, but no audio packets or audioStreamEnd are sent. [R4]	Official guidance says a paused stream should send audioStreamEnd. [S2][S6]	Add VoiceProvider.endAudioStream and invoke once when mute turns on.
5	GoAway ignores timeLeft	The provider logs timeLeft but waits a fixed four-second deadline. [R2]	The server already supplies the remaining connection time.	Parse timeLeft and reconnect before the actual deadline.
6	Transcript fragments become utterances	Every input/output transcription fragment is immediately appended to transcript. [R2][R4]	Fragments can create duplicated or chopped memory input.	Accumulate fragments per speaker and commit on turnComplete/interruption.
7	Output target is not measured	OUTPUT_BUFFER_MS affects requested AudioTrack allocation, but actual effective buffer and underruns are not logged. [R1][R3]	Device minimums can override the target.	Use setBufferSizeInFrames, log actual frames, queue ms, and underruns.
8	Release credential exposure	Gemini key can be compiled into BuildConfig or stored in SharedPreferences, then used in the WebSocket query. [R2][R6][R7]	A long-lived key in a distributed client can be extracted.	Add ephemeral-token mode before public release. [S5]

3.3 What not to change yet

- Do not lower VAD below 500 ms before measuring cut-offs and false turns.
- Do not replace 40 ms capture chunks solely because 20 ms sounds faster on paper.
- Do not remove Base64 from the WebSocket JSON protocol.
- Do not replace automatic VAD with Silero or WebRTC VAD in the first pass.
- Do not add extra phone filters to AudioTrack.
- Do not rewrite the full persona. Improve only the compact delivery block after latency fixes are measured.

4. Implementation plan

Implement in the order below. Each phase has a code contract, test contract, and rollback boundary. The sequence is designed to prevent a blur of simultaneous audio changes that cannot be measured.

P0-A. Decouple playback from the WebSocket callback

Goal: onAgentAudio must enqueue and return quickly. A dedicated thread owns AudioTrack writes. Interruption must atomically clear queued PCM and flush the hardware track.

Files

- Modify: app/src/main/java/com/kenza/callsim/voice/Audiolo.kt
- Modify: app/src/main/java/com/kenza/callsim/call/CallViewModel.kt
- Add: app/src/test/java/com/kenza/callsim/voice/PlaybackQueuePolicyTest.kt

Required design

- PcmPlayer.start creates AudioTrack and a single worker thread with android.os.Process.THREAD_PRIORITY_AUDIO.
- PcmPlayer.enqueue replaces the current blocking write method and returns immediately.
- The worker performs AudioTrack.write(..., WRITE_BLOCKING); blocking is acceptable only on the audio worker.
- Track queued bytes and convert to queued milliseconds using sample rate, mono, and two bytes per sample.
- Keep a bounded low-latency queue. If the queue exceeds 400 ms, record an overflow metric and resynchronize instead of allowing seconds of stale speech.
- flush clears the software queue first, then pause/flush/play the AudioTrack under one lock.
- Log actual bufferSizeInFrames, bufferCapacityInFrames, and underrunCount.
- Use setBufferSizeInFrames after build, because minSdk 26 supports it. Treat the return value as device-dependent.

```
class PcmPlayer(  
    private val sampleRate: Int,  
    private val metrics: PlaybackMetrics = PlaybackMetrics.NOOB,  
) {  
    fun start()  
    fun enqueue(pcm: ByteArray)           // non-blocking caller contract  
    fun queuedDurationMs(): Int  
    fun flush()  
    fun stop()  
}
```

Acceptance check

Instrument the duration of listener.onAgentAudio. Its p95 should remain below 2 ms on a real device, excluding Base64 decoding. No AudioTrack.write call may appear on an OkHttp callback thread.

P0-B. Enable route-aware full duplex

Goal: normal earpiece calls and headset calls must send microphone audio while Gemini is speaking, allowing server-side VAD to interrupt. Speakerphone should remain conservative until echo behavior is proven.

Files

- Modify: app/src/main/java/com/kenza/callsim/voice/Audiolo.kt
- Modify: app/src/main/java/com/kenza/callsim/call/CallViewModel.kt
- Add: app/src/main/java/com/kenza/callsim/voice/AudioRoutePolicy.kt
- Add: app/src/test/java/com/kenza/callsim/voice/AudioRoutePolicyTest.kt

Route policy

Route	Policy	Reason
Wired headset / USB headset / Bluetooth headset	Full duplex	Dedicated mic is separated from speaker.

Route	Policy	Reason
Built-in mic + earpiece	Full duplex	Normal phone route has low acoustic coupling; AEC is already requested.
Built-in mic + speakerphone	Half duplex in first release	Avoid model self-interruption until local double-talk detection is tested.
Unknown route	Half duplex	Safe fallback.

```
object AudioRoutePolicy {
  fun allowsFullDuplex(
    inputDeviceType: Int?,
    speakerphoneOn: Boolean,
  ): Boolean
}
```

- Replace routedInputSupportsBargelIn with a route policy that also receives speakerphone state.
- When full duplex is allowed, do not gate microphone packets on agentSpeaking.
- When full duplex is not allowed, preserve the existing gate.
- Update policy immediately when toggleSpeaker changes.
- Keep AcousticEchoCanceller, NoiseSuppressor, and AutomaticGainControl best-effort. Do not assume they are present.

Rollback switch
Add one internal feature flag that can restore headset-only full duplex if a device family self-interrupts. Do not hide the policy across multiple conditionals.

P0-C. Correct microphone stream lifecycle

Add an explicit provider method for intentional stream pauses.

```
interface VoiceProvider {
  fun endAudioStream() {}
}

override fun endAudioStream() {
  val message = JSONObject().put(
    "realtimeInput",
    JSONObject().put("audioStreamEnd", true)
  )
  socket?.send(message.toString())
}
```

- Call client.endAudioStream exactly once when mute changes from false to true.
- Do not call it repeatedly while muted.
- Unmute requires no special reopen message; the next audio packet reopens the stream.
- Do not call it after each automatic-VAD utterance.

P0-D. Add per-turn telemetry and usage parsing

Telemetry must separate the stages the user feels. It must not store audio or transcript content.

New types

```
data class LiveTurnMetrics(
  val turnId: Long,
  val localSpeechStartMs: Long?,
  val localSpeechEndMs: Long?,
  val lastMicPacketQueuedMs: Long?,
  val firstModelAudioReceivedMs: Long?,
  val firstPlaybackQueuedMs: Long?,
  val firstPlaybackHeadAdvanceMs: Long?,
  val generationCompleteMs: Long?,
  val turnCompleteMs: Long?,
  val interrupted: Boolean,
)
```

```

    val socketQueuePeakBytes: Long,
    val playbackQueuePeakMs: Int,
    val underrunsDelta: Int,
    val totalTokenCount: Int?,
    val route: String,
    val networkClass: String,
    val sessionAgeMs: Long,
)

```

Use a lightweight local speech detector for measurement only. It may use adaptive RMS/noise-floor logic on the existing 40 ms PCM frames. It must not control Gemini turn boundaries in this phase.

- Mark local speech start after two consecutive speech-like frames.
- Mark local speech end after 200 ms of locally quiet frames. This is a metric estimate, not the server VAD rule.
- Record firstModelAudioReceivedMs before calling the listener.
- Record firstPlaybackQueuedMs in PcmPlayer.enqueue.
- Poll playbackHeadPosition briefly to estimate first audible advancement.
- Read WebSocket.queueSize on each send and keep a peak per turn.
- Parse top-level usageMetadata before the provider message-type switch.
- Emit one compact JSON log record when turnComplete arrives.
- Gate persistent telemetry behind a debug/diagnostics setting; never include names or transcript text.

```

json.optJSONObject("usageMetadata")?.let { usage ->
    listener.onUsage(
        total = usage.optInt("totalTokenCount"),
        prompt = usage.optInt("promptTokenCount"),
        response = usage.optInt("responseTokenCount"),
    )
}

```

P1-A. Improve GoAway and reconnection timing

- Parse goAway.timeLeft instead of always using a fixed four-second deadline.
- If the model is not generating, reconnect immediately with the latest resumption handle.
- If it is generating, wait only until min(generationComplete, deadline minus safety margin).
- Use a 500 ms safety margin and clamp malformed timeLeft values.
- Do not play any audio from a superseded connection ID.
- Keep the current maximum internal resume budget, but log each reason and elapsed reconnect time.

P1-B. Assemble transcription by turn

Input and output transcriptions can arrive independently and in fragments. The app should not treat every fragment as a complete utterance.

- Add provider callbacks onGenerationComplete and onTurnComplete with default no-op methods.
- Accumulate user transcription fragments in one buffer for the active user turn.
- Accumulate model transcription fragments in one buffer for the active model turn.
- Commit the assembled text on turnComplete; on interruption, commit only text that was actually delivered and clear the remainder.
- Deduplicate repeated suffix/prefix fragments before appending.
- Keep UI live captions separate from the durable transcript.

P1-C. Tighten prompt and memory budget

The current persona is strong and should not be rewritten. The improvement is to reduce rigid repetition and cap the live briefing more deliberately.

- Keep the permanent persona.
- Change the delivery rule from a universal 3-12 words to conditional response shapes.
- Do not instruct the model to manufacture filler, breaths, or environmental disruptions.
- Cap the dynamic MemoryContext briefing to a tested budget, initially 8,000 characters instead of 14,000.
- Prefer pinned/open-thread/recent-call memories; reduce the count of generic durable memories before truncating profiles.
- Log prompt character count and first-turn time-to-first-audio to validate the cap.

PHONE DELIVERY

- Greetings and acknowledgements: 2-8 words.
- Simple answers: one short sentence.
- Emotional moments: one real reaction, then at most one follow-up.
- Explanations: answer briefly first; add detail only when asked.
- Stop immediately when interrupted.
- Do not manufacture fillers, breaths, or background events.

P1-D. Add ephemeral-token mode for release builds

This is a release security requirement, not a latency experiment. Official guidance says client-to-server Live API apps should use short-lived ephemeral tokens. [S5]

- Keep direct device-to-Gemini WebSocket streaming for latency.
- Add a token broker endpoint that holds the long-lived Gemini API key server-side.
- The broker creates a one-use token constrained to gemini-3.1-flash-live-preview and AUDIO response modality.
- The Android app requests the token over HTTPS before opening the call WebSocket.
- Connect with access_token or Authorization: Token as documented.
- Keep local API-key mode only as an explicitly labeled private developer fallback.
- Remove GEMINI_API_KEY from public release BuildConfig.

5. Tuning plan after the code fixes

Do not tune several variables at once. First establish the new playback and barge-in architecture, then run the matrix below. Product targets are internal goals, not Google guarantees.

Variable	Variants	Measure	Decision rule
VAD silence	500 / 600 / 700 ms	Cut-off rate, response onset, false turn splits	Promote the lowest value with less than 2% clipped/fragmented user turns.
Input chunk	40 ms baseline / 20 ms experiment	Capture-to-send delay, CPU, socket queue, battery	Keep 40 ms unless 20 ms produces a clear real-device improvement.
Playback target	40 / 80 / 120 ms effective buffer	Underruns, server-audio-to-audible delay	Use the smallest stable device-specific target.
Full duplex route	Earpiece / wired / BT / speaker	Self-interruptions, missed barge-ins	Ship earpiece and headsets; gate speakerphone on evidence.
Prompt budget	Current / 8k / 5k dynamic chars	First-turn TTFA, memory recall, style drift	Choose the smallest budget that preserves important continuity.

Internal latency goals

Metric	Target	Reason
Server audio received -> player enqueue	p95 <= 20 ms	Shows receive path is non-blocking.
Local speech end estimate -> first model audio	p50 <= 1,200 ms; p95 <= 2,000 ms	Initial cross-network product target.
Local speech end estimate -> first playback-head movement	p50 <= 1,350 ms; p95 <= 2,200 ms	Closer to user-perceived onset.
Playback queue depth	p95 <= 250 ms	Prevents stale speech.
Playback underruns	< 1 per 5 minutes on stable Wi-Fi	Audio stability.
Barge-in stop	Queued output cleared <= 150 ms after server interruption	Natural overlap behavior.
False self-interruptions	< 1 per 10-minute call	Echo safety.
Clipped user turns	< 2% in test set	Endpoint quality.

Device and environment matrix

Device class	Routes	Network	Conditions
Samsung flagship	Earpiece, speaker, Bluetooth	Wi-Fi and LTE	Quiet room, TV in background, street noise
Pixel-class Android	Earpiece, speaker, wired/USB	Wi-Fi and LTE	Quiet room, music, moving vehicle passenger
Low/mid-range Android	Earpiece, speaker	Wi-Fi and weak LTE	Thermal load, screen on/off, 30-minute call

Conversation test set

- Short greetings with no pause.
- Sentences with a 300-500 ms thinking pause.
- Sentences with a 700 ms pause that should remain one turn.
- User interruption 300 ms after Gemini starts speaking.

- User interruption near the end of Gemini's sentence.
- French and Darija code-switching.
- Whispered speech and low-volume speech.
- Long call with memory references at 5, 15, and 30 minutes.

6. Required tests

6.1 JVM unit tests

Test	Required behavior
GeminiLiveTuningTest	Keep the 20-40 ms and 500-800 ms assertions. Do not narrow to one magic value.
AudioRoutePolicyTest	Headsets and earpiece return full duplex; speakerphone and unknown route return half duplex.
GeminiMessageParserTest	One server event containing multiple parts emits every audio part; usageMetadata is parsed even when another message type is present.
TranscriptAssemblerTest	Fragments merge without duplication; turnComplete commits once; interrupted clears undelivered remainder.
PlaybackQueuePolicyTest	Queue duration math is correct; overflow policy cannot accumulate stale seconds; flush clears all queued bytes.
GoAwayDeadlineTest	Malformed, short, and long timeLeft values are clamped safely.
TurnTelemetryTest	Metrics close once per turn and contain no transcript/audio fields.

6.2 Real-device checks

- Verify onAgentAudio callback timing with Android Studio profiler or timestamp logs.
- Verify no stale model phrase plays after interruption.
- Verify earpiece barge-in without recurrent self-interruption.
- Verify speakerphone does not send echo-triggered user turns in the default conservative mode.
- Verify mute sends one audioStreamEnd and unmute resumes audio packets.
- Verify session resumption continues the same context across the approximately 10-minute connection rollover.
- Verify the 24 kHz AudioTrack is used for Gemini output.

6.3 Build commands

```
./gradlew test
./gradlew assembleDebug
./gradlew :app:testDebugUnitTest
```

6.4 Codex completion report

Codex must finish by reporting:

- Changed files and purpose of each.
- Commands run and results.
- Before/after latency metrics from at least 20 turns.
- Barge-in results by audio route.
- Known device-specific limitations.
- Any recommendation from this document that was not implemented, with a concrete reason.

7. File-by-file implementation checklist

File	Action
voice/GeminiLiveTuning.kt	Keep 40 / 20 / 600 / 100 defaults. Add experimental constants only behind diagnostics. Add queue overflow and telemetry thresholds.
voice/VoiceProvider.kt	Add default endAudioStream, onUsage, onGenerationComplete, and onTurnComplete methods so ElevenLabs remains source-compatible.
voice/GeminiLiveProvider.kt	Parse usage first; emit completion callbacks; add endAudioStream; expose WebSocket queue size; improve GoAway deadline; keep JSON/Base64 and all-part parsing.
voice/AudioIo.kt	Refactor PcmPlayer to enqueue + worker; record actual buffer/underruns; replace headset-only route test with AudioRoutePolicy; keep PCM16 16 kHz capture.
voice/AudioRoutePolicy.kt	New pure policy for earpiece/headset full duplex and speakerphone safe fallback.
voice/LiveTurnTelemetry.kt	New content-free metrics state machine and JSON output.
call/CallViewModel.kt	Use enqueue; update full-duplex policy when speaker changes; call endAudioStream on mute; assemble transcript by turn; connect metrics.
memory/MemoryContext.kt	Reduce dynamic briefing budget after measuring; prioritize recent/open/pinned context.
config/ConfigRepository.kt	Add diagnostics flag and optional token-broker URL. Keep private developer-key fallback clearly labeled.
app/build.gradle.kts	Do not add libraries for first-pass VAD. Remove release GEMINI_API_KEY only when token mode is wired.
README.md / LAUNCH.md	Document latency diagnostics, route behavior, ephemeral tokens, and release security.

Explicit no-op list

Codex should leave these unchanged unless a compile fix is required:

- Compose visuals and call-screen layout.
- Scheduled call behavior.
- ElevenLabs provider network protocol.
- Memory database schema.
- Kenza identity, relationship facts, and core profile.
- Dialing delay and ringtone behavior.

8. Sources and audit references

All external technical claims in this specification were checked against official primary documentation current on July 29, 2026. The original research document was used as an input to audit, not as authority.

ID	Source	Document
S1	Google AI for Developers	Gemini 3.1 Flash Live Preview model page
S2	Google AI for Developers	Live API capabilities guide
S3	Google AI for Developers	Live API best practices
S4	Google AI for Developers	Get started with Live API using WebSockets
S5	Google AI for Developers	Ephemeral tokens
S6	Google AI for Developers	Live API WebSockets reference
S7	Google AI for Developers	Session management with Live API
S8	Android Developers	AudioTrack API reference
S9	Android Developers	AudioRecord API reference

Repository references

ID	Repository path	Audited lines
R1	app/src/main/java/com/kenza/callsim/voice/GeminiLiveTuning.kt	lines 9-20
R2	app/src/main/java/com/kenza/callsim/voice/GeminiLiveProvider.kt	lines 31-43, 128-169, 172-236, 249-302
R3	app/src/main/java/com/kenza/callsim/voice/AudioIo.kt	lines 19-28, 30-167, 191-245
R4	app/src/main/java/com/kenza/callsim/call/CallViewModel.kt	lines 183-240, 340-388, 660-678
R5	app/src/main/java/com/kenza/callsim/voice/VoiceProvider.kt	lines 14-43
R6	app/src/main/java/com/kenza/callsim/config/ConfigRepository.kt	lines 54-76, 138-159, 189-240
R7	app/build.gradle.kts	lines 10-15, 30-42, 97-118
R8	app/src/main/java/com/kenza/callsim/memory/MemoryContext.kt	lines 10-45, 48-127, 217

Source research input

research kenza delay_260729_041530.docx. The source contains 278 pages of repeated latency and realism recommendations. Its valid core is retained here; unsupported, conflicting, obsolete, and non-applicable material has been removed.

9. Paste-ready Codex handoff prompt

Use the following prompt together with this PDF. The PDF remains the detailed source of truth.

You are implementing the attached "Allo Gemini Live Latency, Barge-in and Realism" specification in repository TheHighBrid/Allo at main commit f31166fc3239751b67f99d0bd256193c341e6224.

Create branch codex/gemini-live-latency-v2.

Implement all P0 items, then P1 items in the stated order. Preserve the UI, Kenza identity and core persona, memory schema, scheduled-call behavior, and ElevenLabs behavior. Keep minSdk 26. Use the existing Kotlin, coroutines, and OkHttp stack.

Critical rules:

1. Keep direct WebSocket messages as JSON with Base64 PCM Blob data.
2. Keep 40 ms input chunks and automatic VAD at 600 ms as the initial baseline.
3. Do not add manual endpoint VAD, proactive audio, affective dialogue, fake filler audio, telephony EQ, dummy warm-up input, TCP_NODELAY workarounds, or forced session refresh.
4. Move AudioTrack writes off the OkHttp callback onto a dedicated audio-priority worker with a bounded low-latency queue.
5. Enable full duplex on earpiece and headset routes; keep speakerphone conservative.
6. Add audioStreamEnd only for intentional mic pauses such as mute.
7. Add content-free per-turn telemetry and parse usageMetadata.
8. Improve GoAway deadline handling and assemble transcription fragments by turn.
9. Add optional ephemeral-token mode for distributed builds, while retaining a clearly labeled private developer fallback.
10. Add tests and run ./gradlew test and ./gradlew assembleDebug.

Use small commits matching the sequence in Section 0. At completion, report changed files, test results, before/after metrics from at least 20 turns, route-specific barge-in results, limitations, and any skipped item with reason.

END STATE

The finished app should feel faster because the network callback is never held hostage by the speaker, more human because normal phone routes can interrupt, and easier to improve because every turn produces trustworthy stage-by-stage evidence.